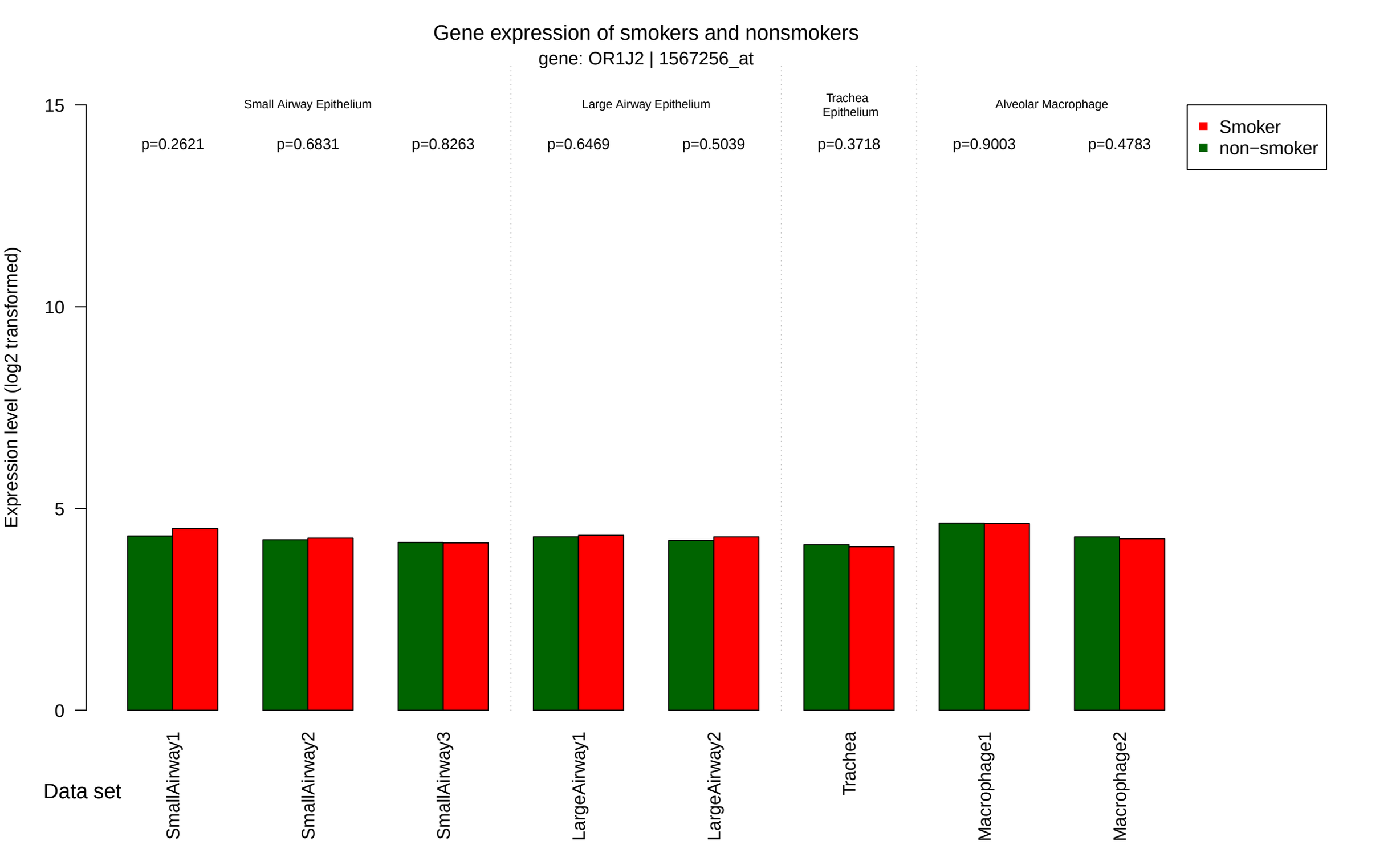
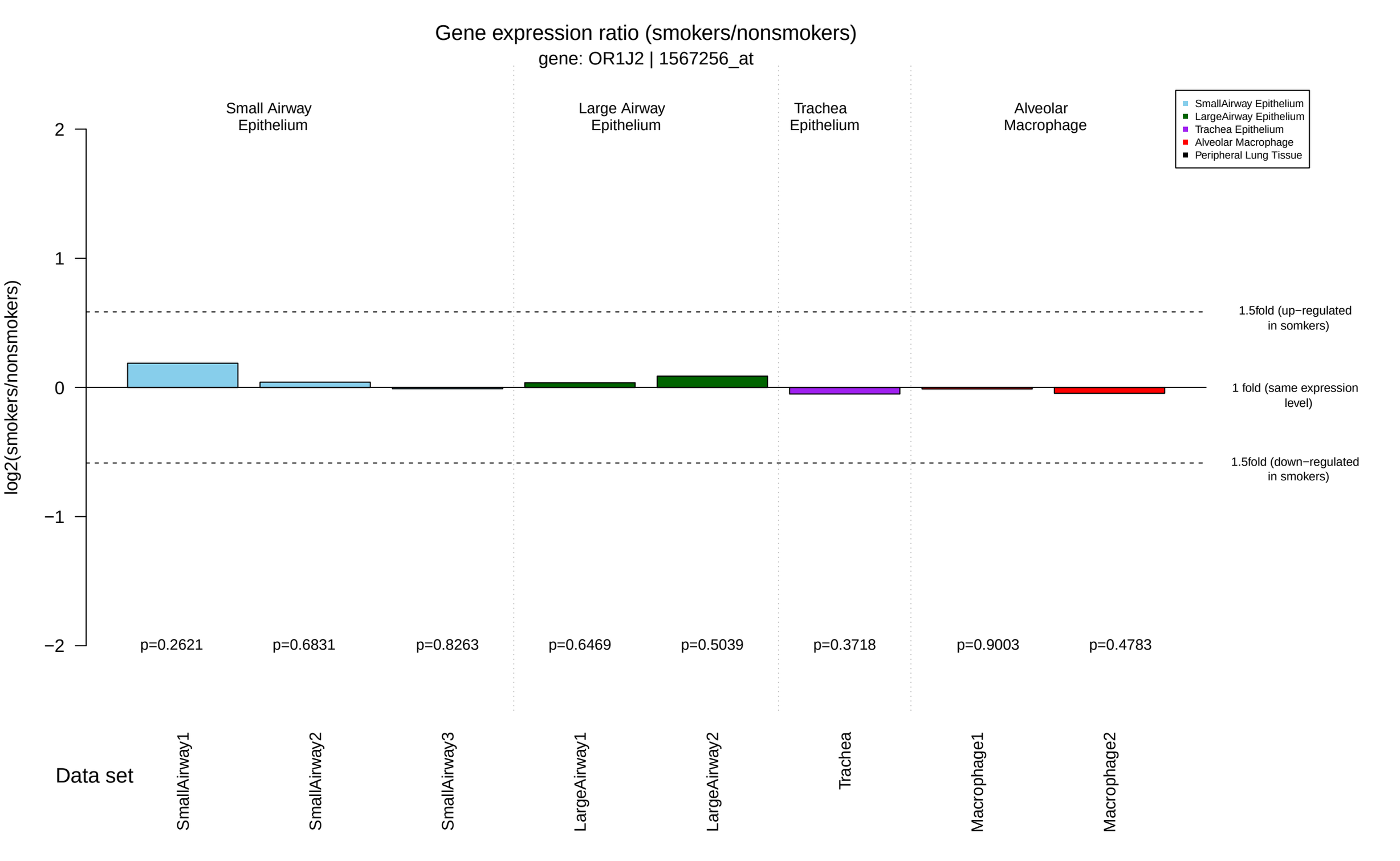
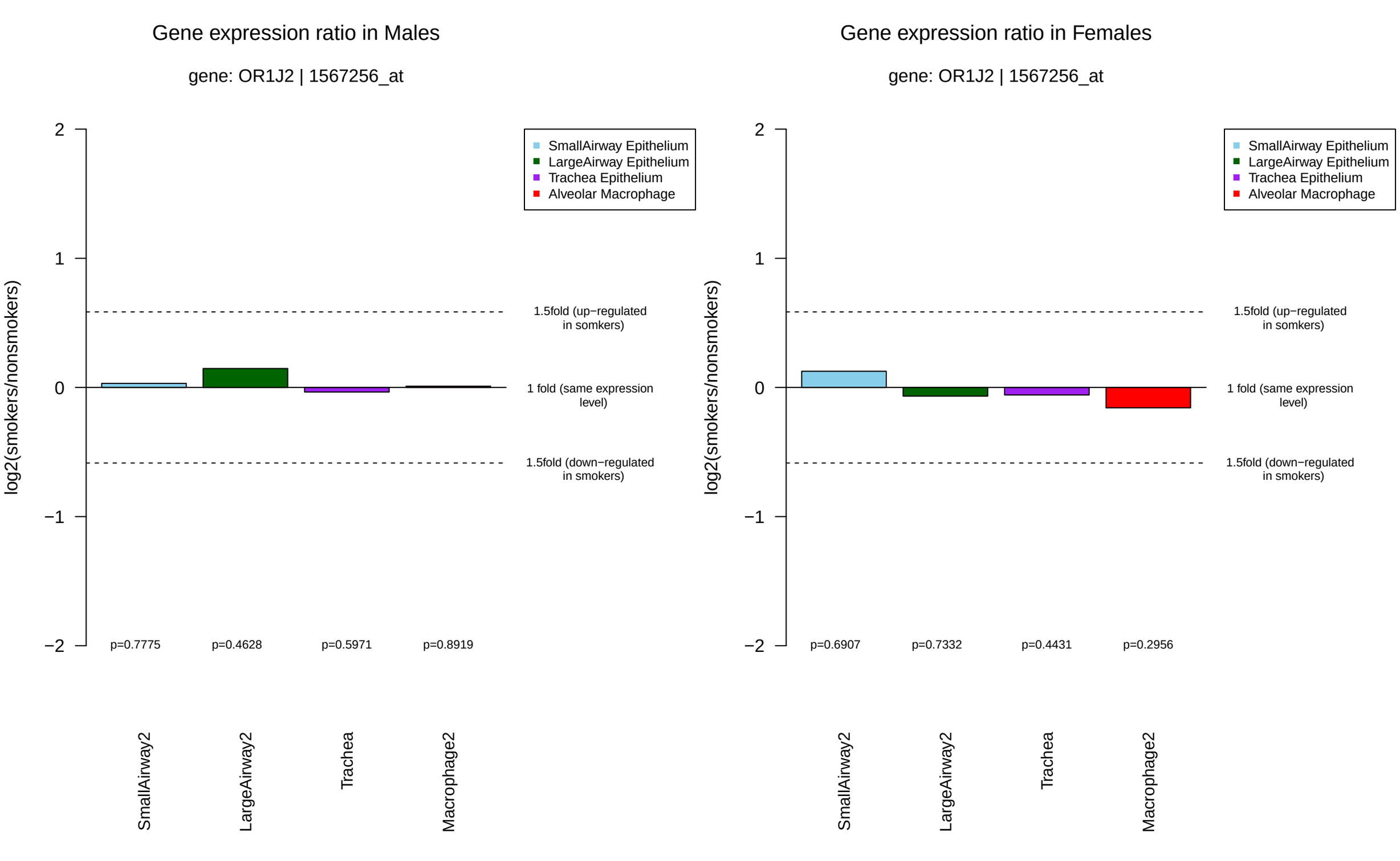
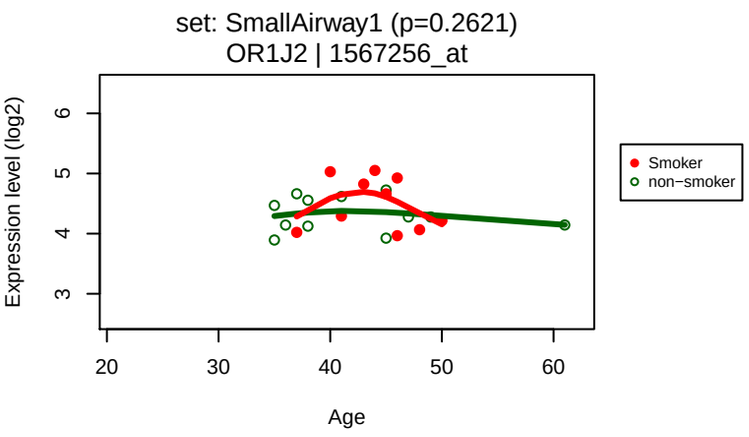
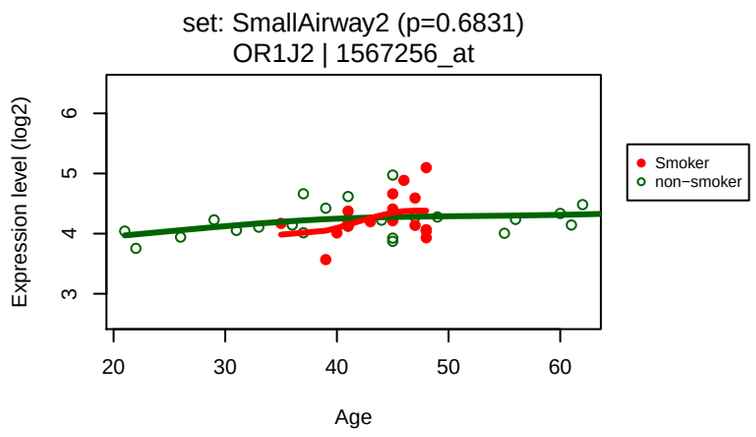
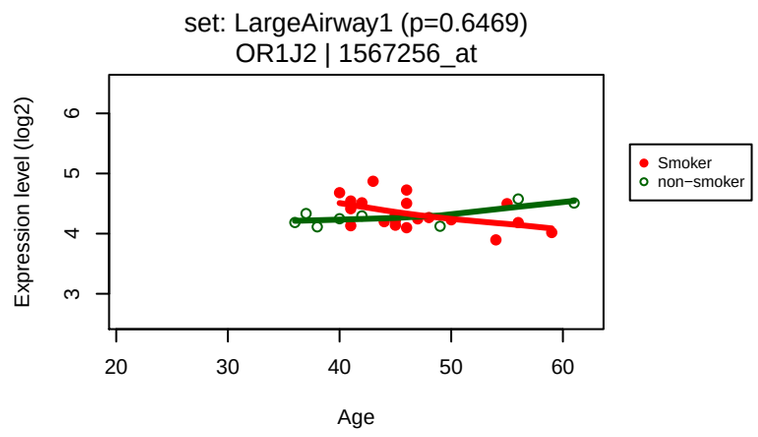

Gene expression of smokers and nonsmokers
gene: OR1J2 | 1567256_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2621
p=0.6831
p=0.8263
p=0.6469
p=0.5039
p=0.3718
p=0.9003
p=0.4783
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

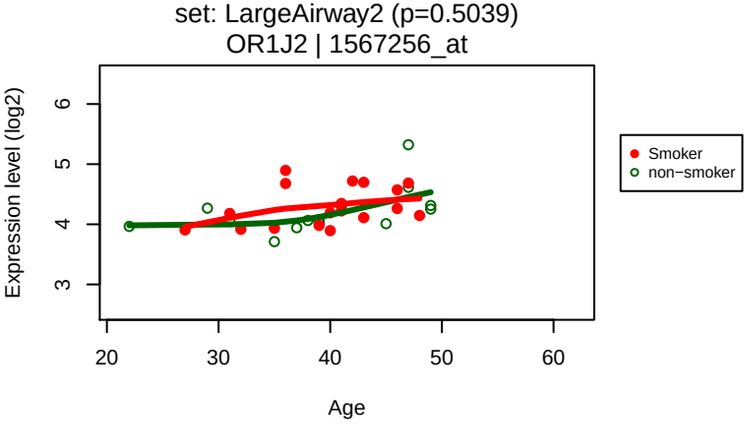

Gene expression ratio (smokers/nonsmokers)
gene: OR1J2 | 1567256_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2621
p=0.6831
p=0.8263
p=0.6469
p=0.5039
p=0.3718
p=0.9003
p=0.4783
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

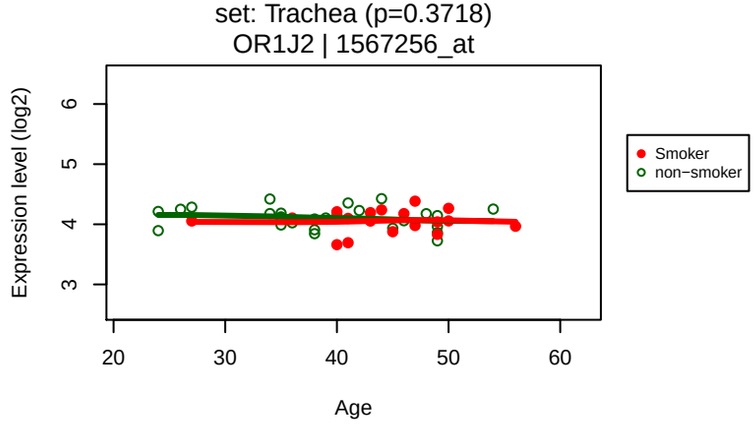

Gene expression ratio in Males
gene: OR1J2 | 1567256_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7775
p=0.4628
p=0.5971
p=0.8919
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR1J2 | 1567256_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6907
p=0.7332
p=0.4431
p=0.2956
SmallAirway2
LargeAirway2
Trachea
Macrophage2

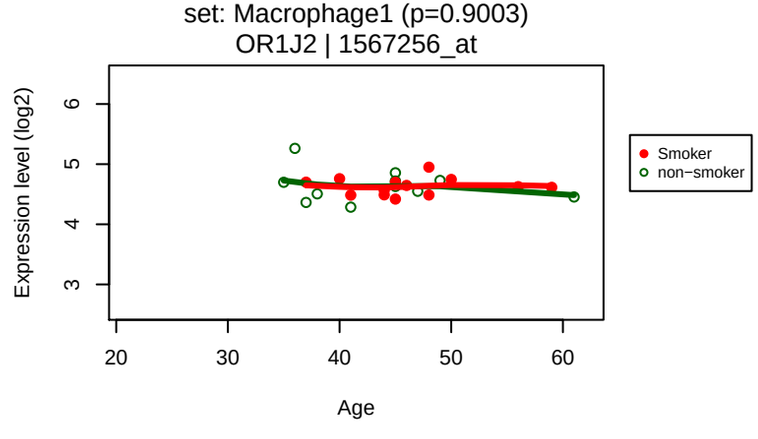

set: SmallAirway1 (p=0.2621)
OR1J2 | 1567256_at
Expression level (log2)
Age
Smoker
non−smoker

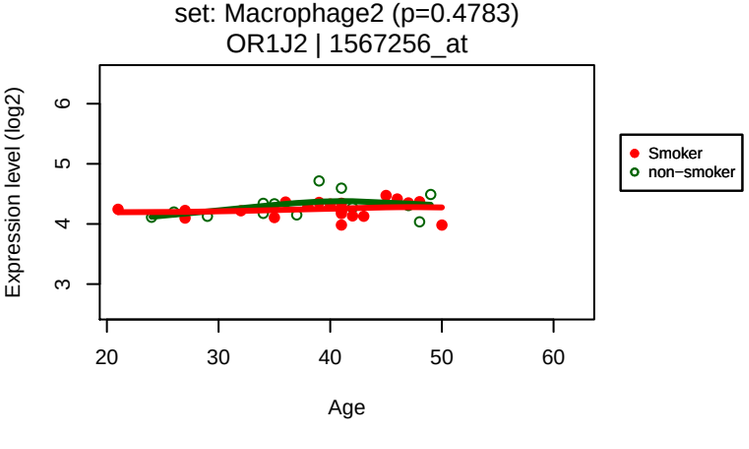

set: SmallAirway2 (p=0.6831)
OR1J2 | 1567256_at
Expression level (log2)
Age
Smoker
non−smoker

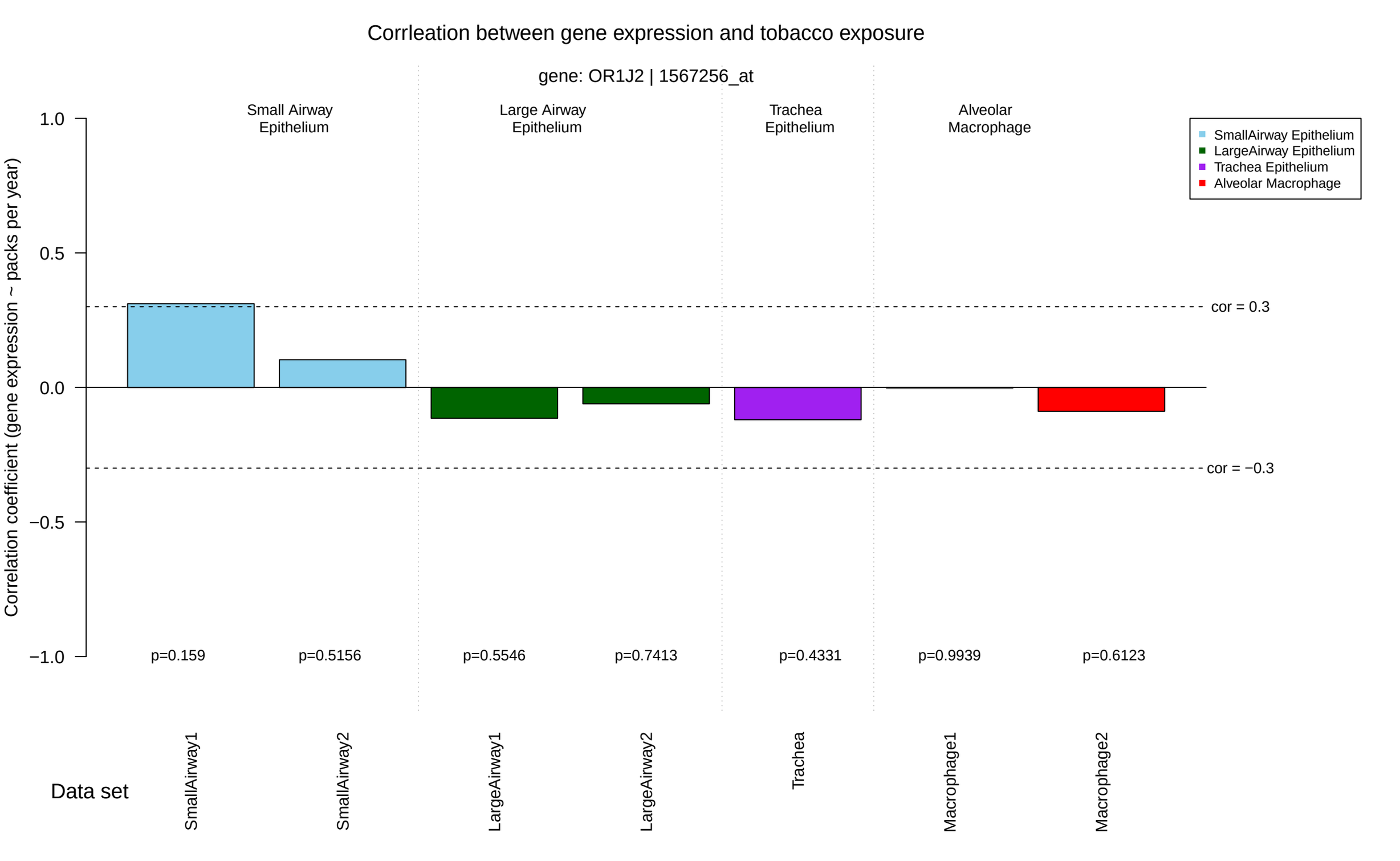

set: LargeAirway1 (p=0.6469)
OR1J2 | 1567256_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5039)
OR1J2 | 1567256_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3718)
OR1J2 | 1567256_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9003)
OR1J2 | 1567256_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4783)
OR1J2 | 1567256_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR1J2 | 1567256_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.159
p=0.5156
p=0.5546
p=0.7413
p=0.4331
p=0.9939
p=0.6123
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2